

Air levitation control system

Stage 2

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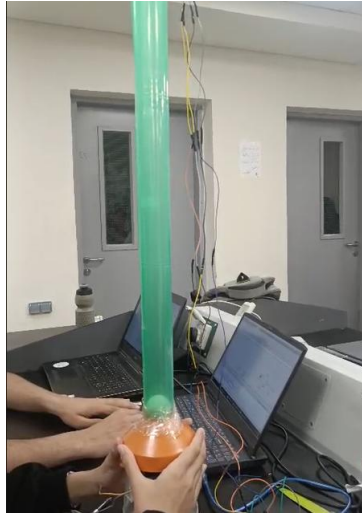
Under the Supervision of:

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Zewail City of Science and Technology
Spring 2026

We have successfully completed our setup and performed the mathematical analysis for this stage.

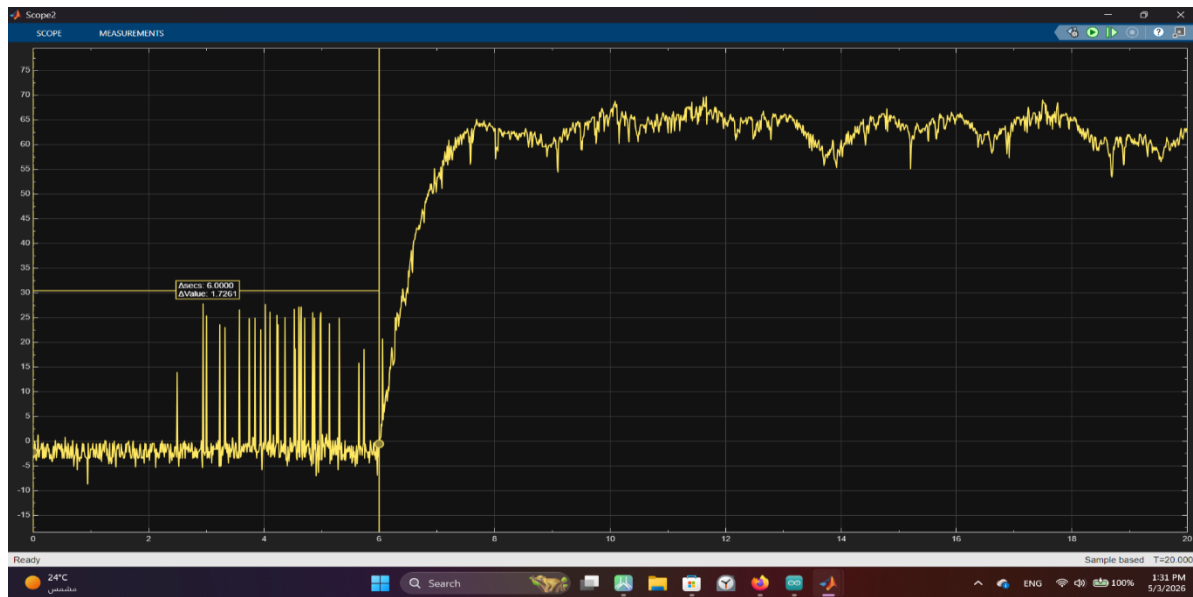
Our completed setup:



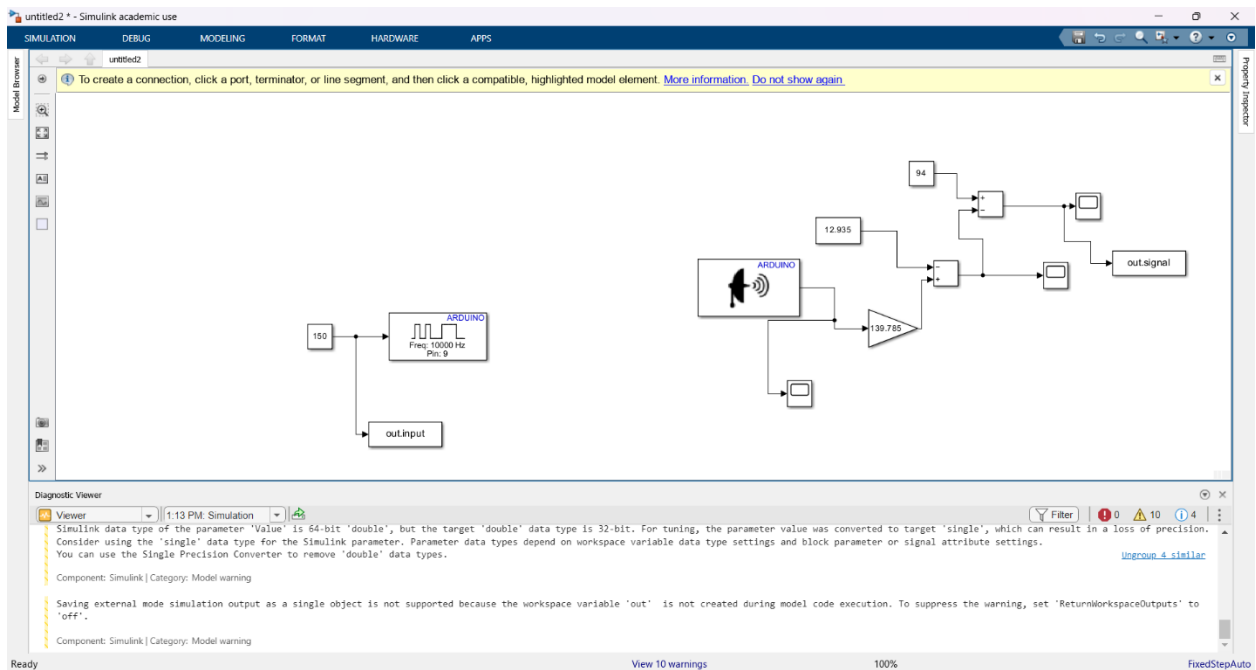
Also we have made a video for our working system:

https://drive.google.com/file/d/11mYf-6kLQ8GkJO9I-CmH1SHdcjRzo1Nq/view?usp=drive_link

System identification and transfer function:

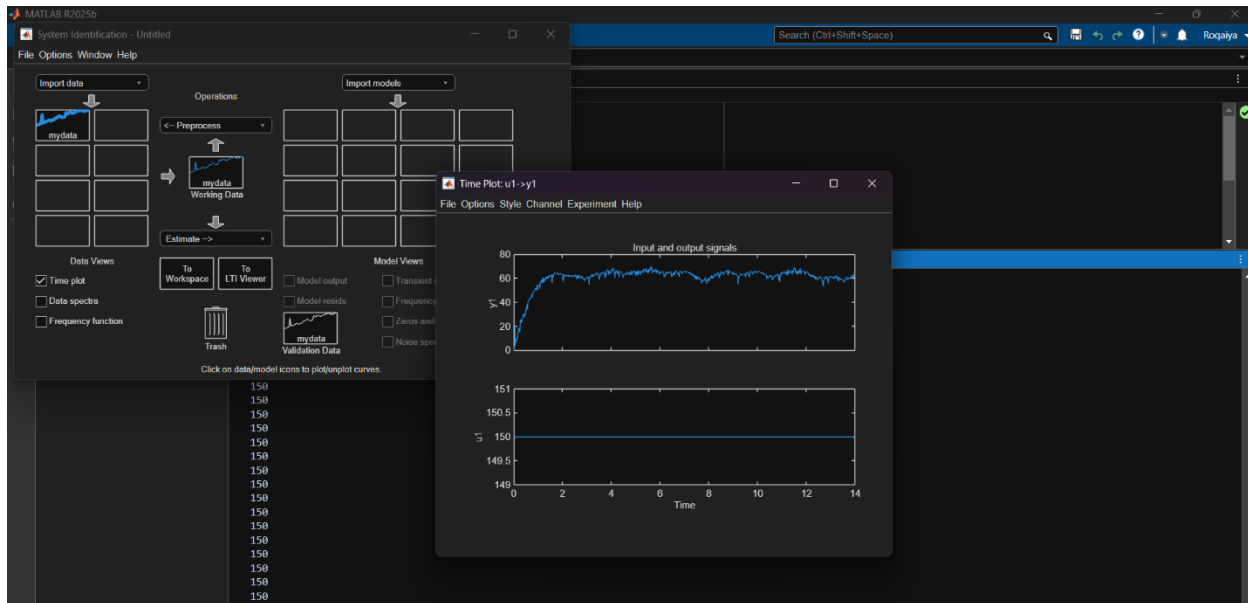


Natural response of the fan under the effect of step input=150 (58.8% of full speed) on PWM pin.

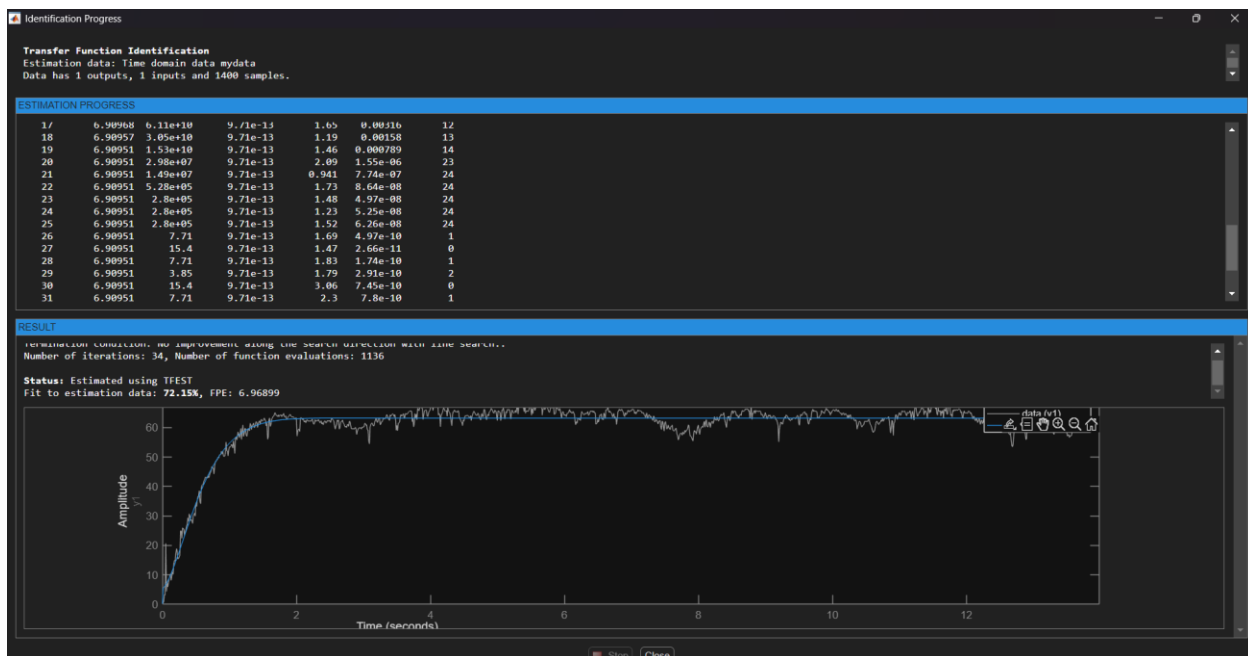


We made some blocks for the output signal to calibrate the distance, as shown.

We had to cut the first 6 seconds of the output (since the ball was not able to rise at this range of voltage, resulting in a noise signal).

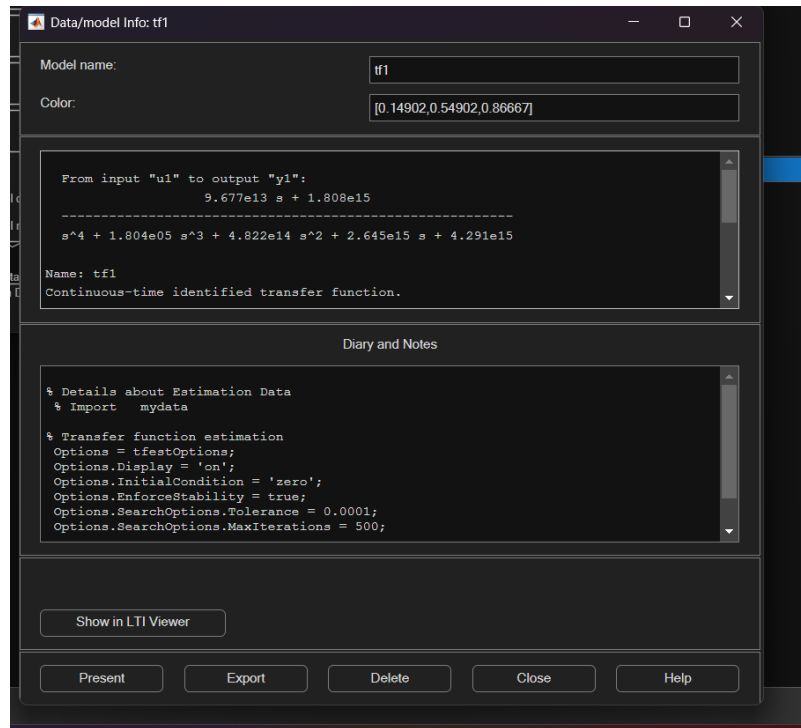


The correct output signal after cutting the undesired first 6 seconds.



Transfer function identification

We have got a 72.15% accuracy!



We guessed the transfer function as the theoretical analysis says, which is in the form of:

$$G(s) = \frac{y(s)}{u(s)} = \frac{b \cdot k_v \cdot (1 - T_d s)}{s(s+b)(\tau s+1)(T_d s+1)} \text{ (4 poles and 1 zero).}$$